

Class 1: Preparation for Learning Edge Networks

XG History and Course Tools

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CIS 4930/5930 Future Edge Networks
<https://xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks/>

Meet Your Instructor



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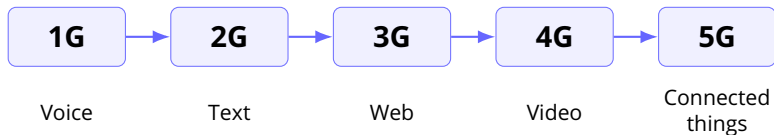
Research:

**AI-driven edge networks and
intelligent wireless systems**

Hobby: *Not Working*
Started at FSU in Fall 2024

[Click here to visit my homepage](#)

Mobile has made a leap every 10 years



1980s 1990s 2000s 2010s 2020s

Each generation changed what people expected a phone to do. The device, radio link, and battery budget moved together.

Play the Nokia ringtone memory.

Every time I hear this tune, I remember my youth, the beautiful days, and the people who made those days bright.

- A familiar sound carried a decade of mobile culture.
- Ringtones made the phone personal.
- Polyphonic audio showed that phones had become small media devices.
 - What makes a computer powerful?
 - Compute throughput, memory capacity, memory bandwidth, storage capacity and speed, I/O bandwidth, processing latency, energy efficiency...



The phone leaves the wall.

- Analog cellular voice
- Expensive devices and sparse coverage
- The core experience: calling from a car or the street

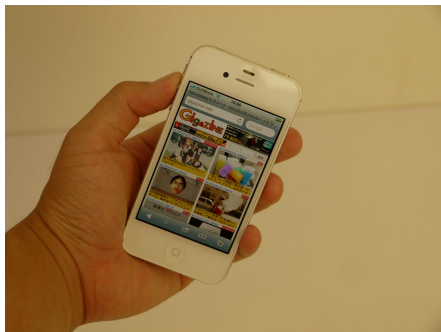
The Motorola DynaTAC “brick phone.”



The phone becomes personal.

- Digital voice and SMS
- SIM cards, contacts, and prepaid plans
- Ringtones, Snake, and the always-on pocket device

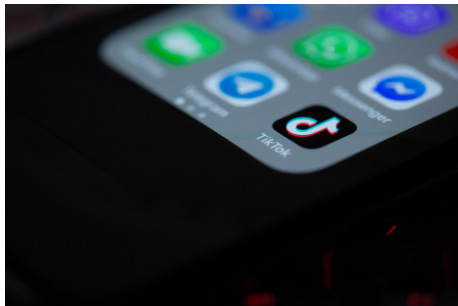
Nokia-style durable feature phones.



A white iPhone 4, a defining device of the 3G era, browsing the mobile web.

Data becomes the product.

- Mobile web, email, and app stores
- Camera phones and multimedia messages
- Early video calls over UMTS networks



TikTok and short-video apps.

The phone becomes a video network.

- LTE makes mobile broadband feel routine
- Streaming, ride sharing, maps, and live apps
- Short video turns uplink and downlink into daily traffic



Active antenna units and dense radio sites.

Connectivity moves into systems.

- Enhanced mobile broadband
- Low-latency control and edge computing
- Sensing, vehicles, factories, and AI services



mmWave 5G often means dense street-level infrastructure.

The route matters.

- Country U had the prime mid-band spectrum tied up. Its carriers could obtain millimeter-wave spectrum more easily.
- Country C chose mid-band coverage first because millimeter-wave small cells cost too much at national scale.
- Country U fell behind in early 5G, but now returns to millimeter wave. Why?

The next “G” should make us ask a different question.

- If every device can run or reach a large model, what should the network provide?
- Which tasks should stay on the device, move to nearby edge compute, or go to the cloud?
- What new applications appear when wireless links carry model state, sensor context, and real-time feedback?



wici.ai/wici-one

Local AI needs nearby compute.

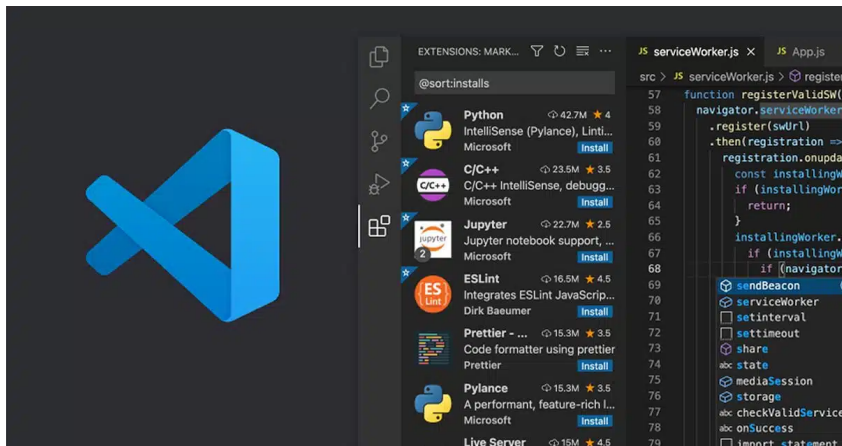
- Phones and laptops cannot carry unlimited GPU, heat, and battery.
- Cloud AI adds latency, cost, privacy risk, and network dependence.
- WiCi One turns nearby GPU into a wireless resource.
- The user keeps a light device; the heavy compute stays nearby.

What else could XG make possible?

Course Tools: IDE, AI, HTML, GitHub,

The Best IDE in the World:

Visual Studio Code

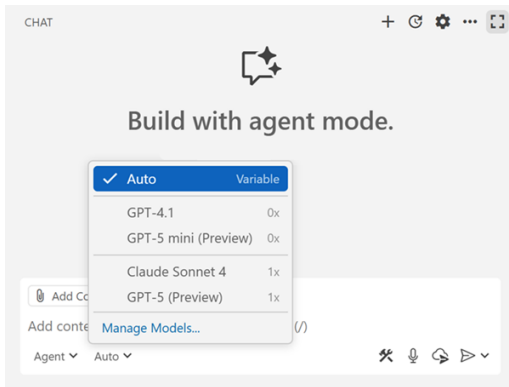


Just remember one thing:

Ctrl + Shift + P

This shortcut opens the Command Palette – your gateway to:
Extensions, settings, themes, terminal, and everything else.

If Ctrl + Shift + P Is Not Enough



Ask an agent clearly

- Give it files, errors, screenshots, and goals
- Then verify: compile, run, and read the diff

VS Code agent mode can work with openAI and Claude models.

My View on AI

Can we use **ChatGPT, Claude, Gemini, Copilot, etc.**
to help with assignments?

Absolutely!



- You're welcome to use AI tools to brainstorm, generate code, or write explanations.
- You can even include notes like “This answer was inspired by ChatGPT 5.6 Sol” — that's fine.
- I want you to learn how to use AI productively, just like in real-world engineering.

- My job is to make sure assignments are designed so that AI won't give you perfect answers easily.
- Your real skill is not in copying the AI's answer—it's in understanding, adapting, and improving what AI gives you.

AI is welcome. Lazy copying is not.

We Use HTML for Assignment Answers

- In the past, JSON was often treated as the standard language for LLMs, APIs, configs, and structured data exchange.
- JSON is still useful, but it is not always the best way to communicate in AI era.

For this course, submit all assignment answers as **HTML files**.

You are expected to use AI tools to help generate your HTML submissions.

- HTML is readable by people and LLMs, and it can be viewed immediately in a web browser.
- It requires no Word, LaTeX, PowerPoint, or specialized viewer.

GitHub Is Our Assignment Interface

- **One interface:** your assignment repository is the submission point.
- **Work in place:** code, write, test, and revise in the repo.
- **Submit by pushing:** your final version lives on GitHub.
- **Real practice:** just like in real-world engineering work.

We may occasionally use Canvas for course logistics, grades, or simple uploads.

- We are living in the **AI era**.
- With AI, the distance between you and “top talent” is much smaller than you think.
- Knowledge is no longer the bottleneck. **Curiosity is.**
- Your advantage is the ability to **ask good questions** and **analyze what you get**.